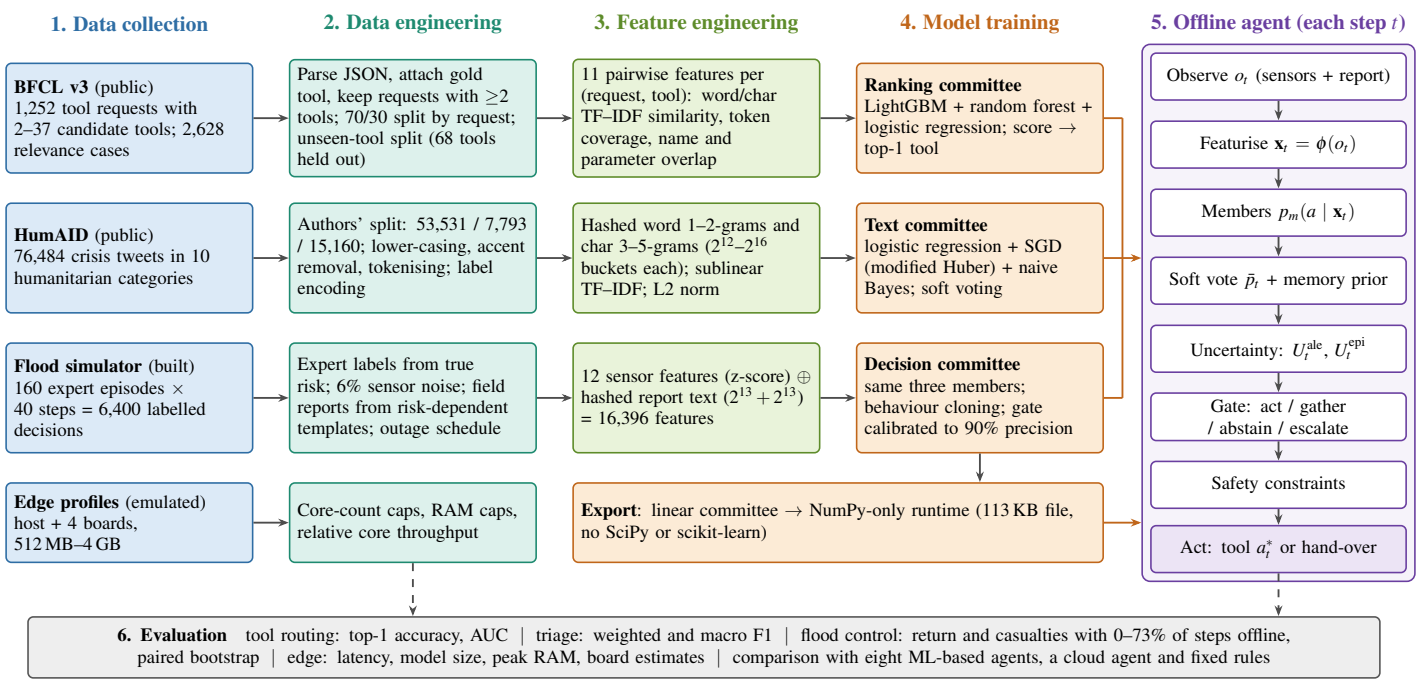

1. Data collection
2. Data engineering
3. Feature engineering
4. Model training
5. Offline agent (each step t)
BFCL v3 (public)
1,252 tool requests with 2–37 candidate tools; 2,628 relevance cases
Parse JSON, attach gold tool, keep requests with ≥2 tools; 70/30 split by request; unseen-tool split (68 tools held out)
11 pairwise features per (request, tool): word/char TF–IDF similarity, token coverage, name and parameter overlap
Ranking committee
LightGBM + random forest + logistic regression; score → top-1 tool
HumAID (public)
76,484 crisis tweets in 10 humanitarian categories
Authors' split: 53,531 / 7,793 / 15,160; lower-casing, accent removal, tokenising; label encoding
Hashed word 1–2-grams and char 3–5-grams (2^{12}–2^{16} buckets each); sublinear TF–IDF; L2 norm
Text committee
logistic regression + SGD (modified Huber) + naive Bayes; soft voting
Flood simulator (built)
160 expert episodes × 40 steps = 6,400 labelled decisions
Expert labels from true risk; 6% sensor noise; field reports from risk-dependent templates; outage schedule
12 sensor features (z-score) ⊕ hashed report text ($2^{13}+2^{13}$) = 16,396 features
Decision committee
same three members; behaviour cloning; gate calibrated to 90% precision
Edge profiles (emulated)
host + 4 boards, 512 MB–4 GB
Core-count caps, RAM caps, relative core throughput
Export: linear committee → NumPy-only runtime (113 KB file, no SciPy or scikit-learn)
Observe o_t (sensors + report)
Featurise $\mathbf{x}_t = \phi(o_t)$
Members $p_m(a \mid \mathbf{x}_t)$
Soft vote $\bar{p}_t$ + memory prior
Uncertainty: U_t^{ale}, U_t^{epi}
Gate: act / gather / abstain / escalate
Safety constraints
Act: tool a_t^* or hand-over
6. Evaluation tool routing: top-1 accuracy, AUC | triage: weighted and macro F1 | flood control: return and casualties with 0–73% of steps offline, paired bootstrap | edge: latency, model size, peak RAM, board estimates | comparison with eight ML-based agents, a cloud agent and fixed rules